

AN ELLIPSE FITTED TRAINING-LESS MODEL FOR PEDESTRIAN DETECTION

Arindam Sikdar Ananda S. Chowdhury

Department of Electronics and Telecommunication Engineering
Jadavpur University, Kolkata - 700032, India.

E-mails: arindamsikdarju@gmail.com, aschowdhury@etce.jdvu.ac.in

ABSTRACT

The problem of pedestrian detection has gained much popularity in the computer vision community in recent times. We have noted that the existing solutions to this problem are mostly supervised in nature. However, it is difficult to guarantee availability of labelled training data in all situations. In this paper, we propose a training-less solution of pedestrian detection. Some of the additional challenges for pedestrian detection are proper handling of viewpoint dependencies, background clutter, illumination variation and occlusion. We design an ellipse fitting model, as a part of our training-less solution, for accurate pedestrian detection. In this model, we fit an ellipse to each competing bounding box (proposal). An area and entropy based quality factor is introduced for every such (fitted) ellipse to discriminate among the proposals. We filter out proposals with low quality factors. Performance comparisons with some well-known supervised pedestrian detection approaches on publicly available PETS2009 dataset demonstrate that our solution is highly promising.

Index Terms—Pedestrian detection, Training-less solution, Ellipse fitting, Quality factor.

I. INTRODUCTION

Pedestrian detection is an important problem in several computer vision tasks like visual tracking, scene understanding and activity recognition. Although several methods on pedestrian detection have been developed in recent years [1]–[5], there still remain many challenges like viewpoint dependencies, background clutter, illumination variation, different possible articulations and multiple occlusion handling in crowded scenes. Moreover, the reported algorithms require varying amount of training data for their successful execution.

This fact motivated us to develop a training-less model of pedestrian detection that does not require any prior training. To the best of our knowledge, there is no dedicated pedestrian detection model in the literature which can work successfully without any form of training/supervision. In this paper, we further develop an ellipse-fitting model for

correctly capturing the pedestrians (within bounding boxes) in the edge-fused foreground map.

II. RELATED WORKS

Many approaches [3], [4] have been developed for detecting pedestrians in several forms of crowd scenes [5]. These methods can be broadly categorized into 3 main categories, namely: a) Sliding window approach [6]–[8], b) Part-based approach [9]–[11], and c) Patch based approach [12]–[14].

In the sliding window approach, detectors are trained based on features obtained from available full-body pedestrian regions collected over several dataset and a classifier such as Support Vector Machine (SVM) is trained based on the feature extracted from that set. Then a sliding window is moved over the full frame and the detector pops-up a window if the features from the sliding window match with that of the classifier. Features like SIFT [6] HOG [7] and LBP [8] are mostly adopted for training the classifier. While the above approach mostly used gradient based features, part-based models used several human body parts like, head, torso, leg to train the detectors. They have mostly used local shape features like edgelet and shapelet for training models like DPM [10], [11]. After training the model pedestrians are detected by assembling body-part hypothesis in the detection frame. Patch-based methods achieved detection mostly by segmenting regions into several appearance patches and extracting key-entries to learn a code-book during the training process. For example, see the Implicit shape model (ISM) by Leibe *et al.* [12]. Key-points are extracted from detection regions based on the match over the entries of the code-book and are assigned a score based on the number of matches. These approaches, though require less training data, do not perform very well in low resolution videos.

The above methods suffer from an universal limitation and that is the necessity of training. Since, feature extraction in low resolution videos becomes difficult, the performance of the above detection methods also gets affected in such cases. We propose a completely “**training-less**” model for pedestrian detection that integrates spatial edge information over temporally generated map. Some additional challenges for pedestrian detection include proper handling of viewpoint

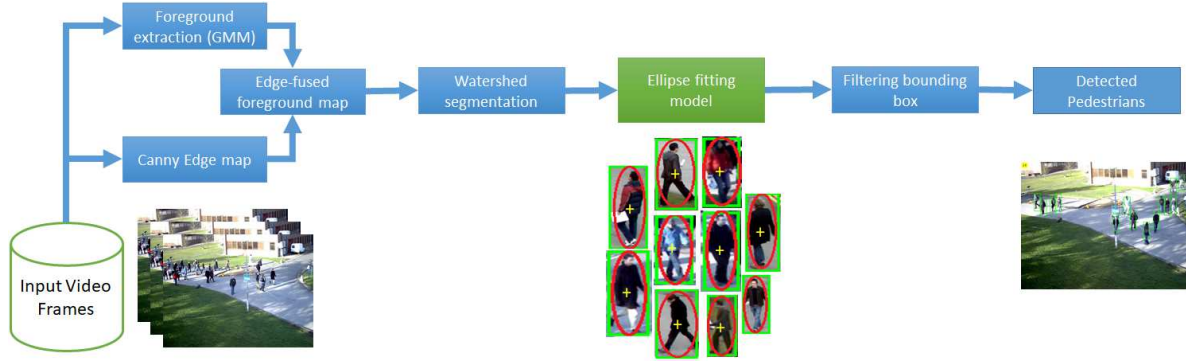


Fig. 1. Different stages of our proposed pipeline.

dependencies, background clutter, illumination variation and occlusion. An ellipse fitting model, as a part of our training-less solution, is applied for accurate pedestrian detection. In this model, we fit an ellipse to each competing bounding box containing a pedestrian (proposal). An area and entropy based quality factor is introduced for every such (fitted) ellipse to discriminate among the proposals. We filter out proposals with low quality factors. An outline of our model pipeline model is described in Fig. 1.

III. PROPOSED FRAMEWORK

Our model consists of three main steps. These are: i) foreground region extraction using Gaussian Mixture Model (GMM), ii) Canny edge-guided Watershed segmentation, iii) Ellipse fitting based quality factor computation and filtering. Outputs from different stages of our model is presented in Fig. 2. Presently, our model works well on videos captured with static camera.

III-A. Foreground Region Extraction using GMM

At first moving object regions from a video frame $I(x, y)$ are determined using the well-known parametric background subtraction technique based on GMM. Under this model, each pixel of the background is drawn from a set of Gaussian probability distributions $P(I_t = u)$ [15] as:

$$P(I_t = u) = \sum_{i=1}^K \omega_{i,t} * \eta(u; \mu_{i,t}, \sigma_{i,t}) \quad (1)$$

Here, $\eta(u; \mu_{i,t}, \sigma_{i,t})$ is the i^{th} Gaussian component at frame t with mean intensity $\mu_{i,t}$ and standard deviation $\sigma_{i,t}$ weighted by $\omega_{i,t}$. All the parameters of GMM are initiated and updated on-the-fly [15] as we sequentially process through each video frames. This eliminates the requirement of separate training for GMM model. A pixel I_t is declared as a foreground pixel if for at least one Gaussian component $\hat{i}$, it satisfies the equation $|I_t - \mu_{\hat{i},t}| \leq D \cdot \sigma_{\hat{i},t}$ where D is a constant. Then, a (5×5) 2D Laplacian of Gaussian (LoG) filter is applied to better capture the edge regions of the foreground objects. The output image $LG(x, y)$ is given by:

$$LG(x, y) = \begin{cases} 1 & \text{if } \frac{\delta^2[I(x, y)]}{\delta x^2} + \frac{\delta^2[I(x, y)]}{\delta y^2} \neq 0 \\ 0 & \text{if } \frac{\delta^2[I(x, y)]}{\delta x^2} + \frac{\delta^2[I(x, y)]}{\delta y^2} = 0 \end{cases} \quad (2)$$

This operation also enhances the discrete noisy regions of the background which are then removed using blob filtering. The filtered image $L_{refined}(x, y)$ thus obtained from $LG(x, y)$ consists of binary regions that better contains the persons.

III-B. Canny-Edge guided Watershed Segmentation

Watershed segmentation algorithm [16] is based on image topography. However, it cannot efficiently segregate regions when two objects share significant overlap. To circumvent this problem, we fused edge information over the foreground region $L_{refined}(x, y)$ to correctly guide the segmentation process through object boundaries. Canny edge map $C_{edge}(x, y)$ is used for this purpose. Thus an edge-embedded foreground region is generated based on following equation:

$$L_{edge-guided}(x, y) = L_{refined}(x, y) \wedge \overline{C_{edge}(x, y)} \quad (3)$$

where, $\wedge$ is the bit-wise AND operation and $\overline{(\cdot)}$ is complementary operation at each pixel (x, y) . See Fig. 2(c) in this connection. For watershed segmentation, a topographical image is generated using distance transform over $L_{edge-guided}(x, y)$ using city-block distance. This is given by:

$$\nabla(L_{edge-guided}) = \left\{ \min_q \|p - q\|; \overline{L_{edge-guided}}(q) = 0 \right\} \quad (4)$$

where, pixels $p, q \in \overline{L_{edge-guided}}$. Since object shape segmentation is considered here, distance transform is used to generate the topographic image [16]. The watershed output labels are computed over $\nabla(L_{edge-guided})$ in the following manner:

$$L = \{w_i : w_i \subset \nabla(L_{edge-guided}(x, y))\} \quad (5)$$

where, w_i is the i^{th} watershed segmented region obtained using [17]. L is the label set assigned to each of the obtained w regions. A centroid set C for each of the w

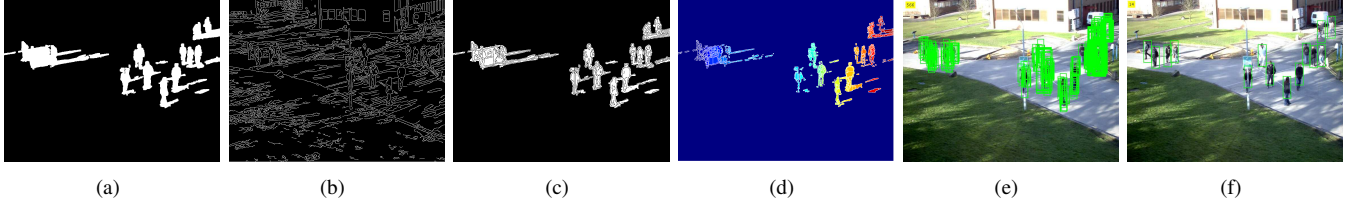


Fig. 2. Output at different stages of our proposed pipeline, where, (a) GMM based extracted foreground; (b) Canny edge map; (c) Canny edge-fused foreground map; (d) Watershed segmented label map; (e) Proposal bounding box based on segmentation; (f) Detected pedestrians.

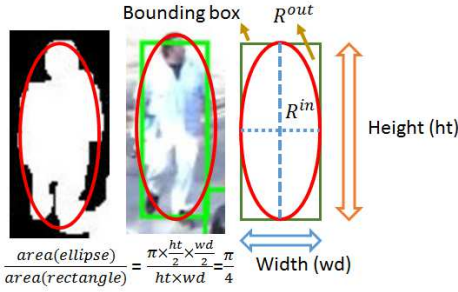


Fig. 3. A Schematic diagram showing our ellipse fitting model in evaluating the area factor ψ .

regions is determined. To find appropriate pedestrian region, several (proposal) rectangular boxes (say N_B) of varying dimensions are generated across these centroid set C . These rectangular boxes are termed as bounding boxes (BB) which potentially contain the pedestrians. Moreover, as stated by Dollar *et al.* [18] the typical aspect ratio of the person is ~ 0.41 which is set as a constraint while generating proposal bounding boxes. Note that the boundaries of the canny edge map is not necessarily required to be closed. It just ensures the segmentation process through edges rather than generating randomized boundary specially at the conjunction of two nearby persons. Also note that multiple segments are generated for same person during segmentation so out of multiple bounding boxes one among them better localizes a person which we tend to extract based on ellipse fitting model as described in next section.

III-C. Ellipse Fitting Model

The proposal bounding boxes at previous stage also contains certain non-pedestrian regions as outliers which need to be removed. We achieve this task by filtering such bounding boxes based on our proposed quality factor computed for each box. The quality factor is composed of two terms, namely, area factor and entropy factor. The area factor is based on the hypothesis that an ellipse can be used to model the boundary of a pedestrian within a bounding box. So, for each proposal box, we geometrically inscribe (best fit) an ellipse as shown in Fig. 3. A bounding box containing

Algorithm 1 Ellipse fitting based pedestrian detection algorithm

Require: Video Sequence $\{frame_t\}$

Ensure: $BB_t \in frame_t$

Initialization: $Q_{thres} \leftarrow 0.75$

- 1: **for** $t = 1$ **to** T **do** /* T is the total no. of frames */
- 2: Obtain foreground region $I(x,y)$ using GMM
- 3: Obtain $L_{refined}(x,y)$ by applying size filtering
- 4: Obtain $L_{edge-guided}(x,y)$ by using Canny-Edge Map $C_{edge}(x,y)$
- 5: Obtain $\#L$ regions using watershed
- 6: Compute centroid set C over $\#L$ segments
- 7: Generate candidate bounding boxes over set C
- 8: Compute quality score Q^{BB} for each candidate bbox based on equation 7
- 9: Filter bounding boxes based on $Q^{BB} > Q_{thres}$
- 10: Apply NMS over filtered detection
- 11: Compute quality score Q^{BB} for previous boxes (BB_{t-1}^d) in current frame t using equation 7.
- 12: Filter bounding boxes based on $Q^{BB} > Q_{thres}$
- 13: Combine the resulting bounding boxes of current frame with the previous (BB_{t-1}^d).
- 14: Apply NMS suppression to obtain final detection (BB_t^d)
- 15: **end for**

pedestrian would be deemed optimal if his maximum portion (*i.e.*, foreground region $f \in L_{edge-guided}$) lies within the fitted ellipse (R^{in}) while most of his surrounding background lies outside (R^{out}) the ellipse within a bounding box. Thus, the area factor ψ_i^{BB} is defined based on a weighted combination of the above two considerations in the following manner:

$$\psi_i^{BB} = \frac{\text{area}(R_i^{in} \cap f)}{\text{area}(R_i^{in})} * \omega + \frac{\text{area}(R_i^{out} \cap \bar{f})}{\text{area}(R_i^{out})} * (1 - \omega) \quad (6)$$

where, ψ_i^{BB} is the area factor for the i^{th} bounding box, $i = 1, \dots, N_B$. Here, $(R_i^{in} \cap f)$ denotes how much foreground region is contained inside the best fitted ellipse, $(R_i^{out} \cap \bar{f})$ denotes how much background region is contained in the bounding box minus the best fitted ellipse, and, ω is the weight factor. It is a fact that the best fitted ellipse holds

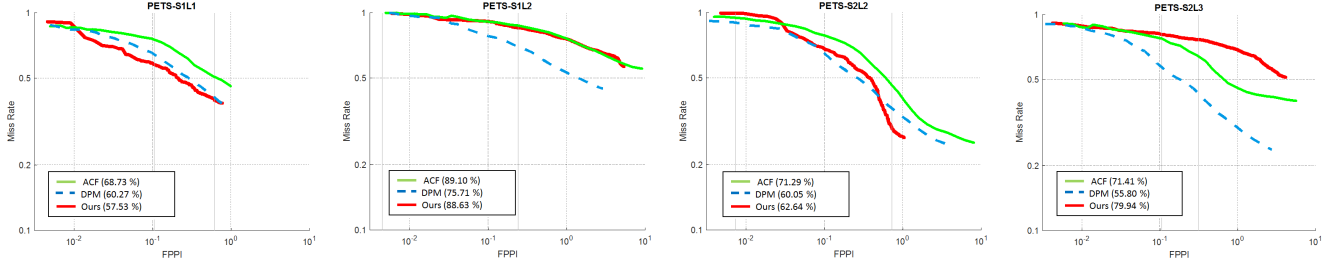


Fig. 4. Detection error tradeoff curves on PETS2009 dataset for different detectors. The x -axis represents false positive per image (FPPI) and y -axis represents the miss rate. The lower the curve the better the performance.

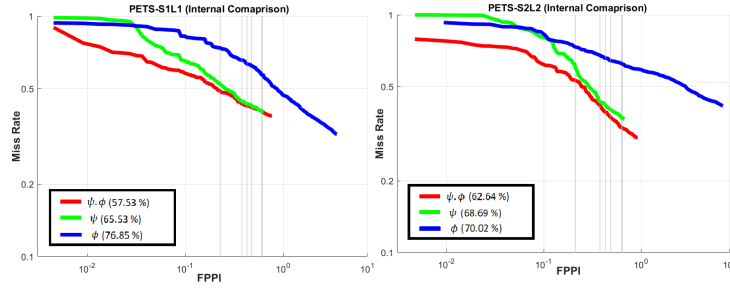


Fig. 5. Performance comparison on PETS2009 dataset showing the effect of each factor (ψ , ϕ) in equation 7 along with the combined effect of both of them.

an area ratio of $\pi/4$ with its rectangular bounding box *i.e.*, $\frac{\text{area(ellipse)}}{\text{area(rectangle)}} = \frac{\pi}{4} \simeq 0.7854$. Thus the permissible range for the values of weight ω is $[\pi/4, 1)$. Any value of $\omega < \pi/4$ will give undue dominance to the second term in equation (6). Within the range of $[\pi/4, 1)$, we experimentally set the value of ω as 0.95. We further introduce a second factor in terms of normalized Shannon entropy to boost the score of a correct bounding box. It has been observed that the bounding boxes which capture major portions of a pedestrian with parts of background yield relatively higher variation of intensities as compared to the boxes containing mostly background. So, the quality factor Q_i^{BB} of each i^{th} bounding box is determined by the product of the area factor and the Shannon entropy and is given by:

$$Q_i^{BB} = \psi_i^{BB} \cdot \phi_i^{BB} \quad (7)$$

In the above equation, $\phi_i^{BB} = -(\sum_k p_{i,k} \cdot \log_2(p_{i,k})) / \log_2(N)$ is the normalized Shannon entropy where $p_{i,k}$ denotes the probability of occurrence of the k^{th} intensity level ($k = 1, \dots, N$) within the i^{th} bounding box with a total of N intensity levels. We denote detected bounding boxes at time t by BB_t^d . In Fig. 6, we have indicated how the area and the entropy factor jointly help determining a more appropriate bounding box in comparison with the ground truth. After computing the Q^{BB} value for each bounding box, we keep only those whose Q^{BB} values are above a threshold Q_{thres} . We experimentally set the value of Q_{thres} to be 0.75. We have applied this filtering in two stages. In the first stage,

the operation is being performed on the proposals of the current frame. For the second stage, we also take into account the final detection in the last frame and combine them with the resulting bounding boxes of the current frame. This is done in order to reduce the miss rate which can accrue if some pedestrians are not detected in the current frame. To suppress multiple response of the same pedestrian, we apply Non-maximal suppression (NMS) after quality factor based filtering. A high quality score ensures more appropriate localization of a pedestrian and kept among the other overlapping bounding boxes that exceeds a certain amount of overlap. If any two bounding boxes is below a certain overlap then both are kept assumed to contain one pedestrian each. The overlap criterion for suppression is based on the PASCAL 2012 detection criteria [4]. A minimum overlap threshold of 10% is chosen in this work. Choice of such threshold is based on the fact that higher threshold values tend to generate more false positives while lower threshold values lead to higher miss rates. The overall strategy of our pedestrian detection method is described in algorithm 1.

IV. EXPERIMENTAL DETAILS

For comparative evaluation, we used PETS2009 [19] dataset. In particular, S1L1, S1L2, S2L2 and S2L3 sequence are taken and the results are compared with the reported methods as given in table I. Out of these four sequences, S1L1 and S2L2 correspond to medium crowd density and

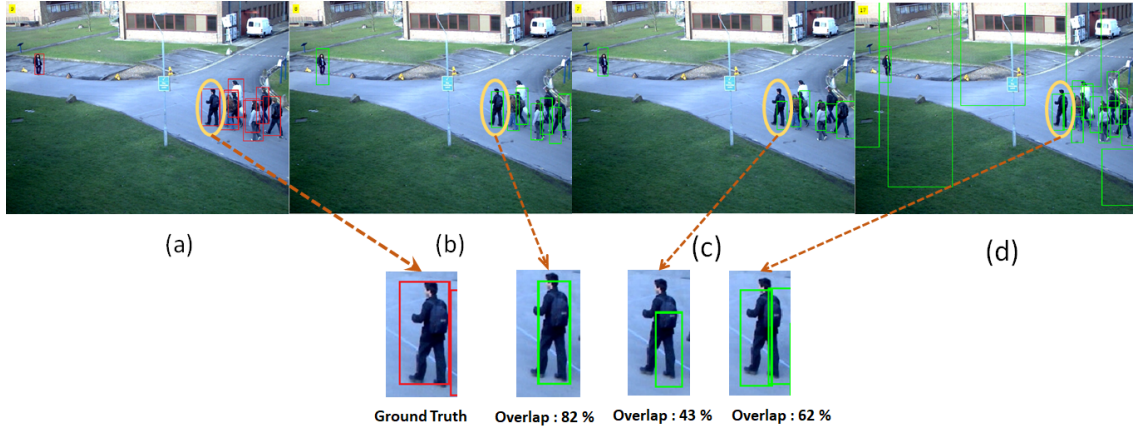


Fig. 6. Comparing the detection performance on a particular instant of PETS2009 S1L1 dataset based on the factors (b) $Q = \psi \cdot \phi$, (c) ψ only, (d) ϕ only and compared its detection overlap with the ground truth (a) based on PASCAL 2012 overlap criterion [4].



Fig. 7. Detection results on various scenes of PETS2009 dataset

Table I. Comparison of our detection result with various methods reported in [4] by setting FPPI to 0.1. (TP, FP, FN, Accuracy = $TP/(TP+FP+FN)$ and Precision = $TP/(TP+FP)$)

Methods	TP	FP	FN	Accuracy	Precision
DPM+NMS [9]	1017	19	3348	23.2	98.2
DPM+DPP [4]	1836	19	2529	41.9	99
ACF+NMS [18]	633	19	3732	14.4	97.1
ACF+DPP [4]	967	19	3398	22.1	98.1
RCNN+NMS [20]	69	19	4296	1.6	78.4
RCNN+DPP [4]	137	19	4228	3.1	87.8
Ours	2053	22	2909	41.2	98.9

S1L2 and S2L3 correspond to high crowd density. All the implementations were done on MATLAB 2016a version on a 64-bit operating system running at a 2.6 GHz clock speed. We employed Log-Average Miss Rate (LAMR) [1] for evaluating our detector performance over PETS dataset

while comparing with the baseline DPM [10] and the state-of-the-art ACF [18] detector as described in [11]. The ROC curves with miss rate versus False positive per Image (FPPI) for the three methods are shown in Fig. 4. From the above plot, it can be inferred that our detector have achieved a miss rate of 57.53% on S1L1 which is better than both ACF and DPM; and 62.64% on S2L2 which is better than ACF and only slightly less than DPM. On the high crowd density sequence S1L2 and S2L3 our method is fairly close to ACF with a miss rate of 88.63% and 79.94%. A possible reason for a relatively high miss rate in these high density crowd sequence is due to heavy occlusion, where visibility of most persons is less than 50%. Although a recent work of [11] has reported superior results compared to ours, that is supervised in nature and has in addition a dedicated occlusion handling model. Another recent work [4] also compared PETS2009 dataset over different variations in methods like ACF, DPM,

and, RCNN in terms of True Positive (TP), False Positives (FP), False negatives (FN), Accuracy and Precision by setting FPPI at 0.1. We also compared our framework with these methods and the results are shown in Table. I. We achieved much higher TP and much less FN compared to all these methods. Additionally, our accuracy is comparable with the highest accuracy achieved by DPM+DPP (the best among the six reported methods). Though our precision is slightly lower than some of the methods due to high False positive, it is significantly better than that of RCNN+NMS [20] and RCNN+DPP [4]. Along with that we also compared the detection performance by considering each factor of the equation 7, *i.e.*, ψ and ϕ separately and together and shown that a joint performance of these two factors gives lower miss rate as shown in Fig. 5. We also computed computational time on different resolutions of video. Our algorithm takes an average of 1.809 sec, 2.982 sec and 9.691 sec per frame on a video of resolutions 238×158 , 360×240 , and 768×576 respectively which is at par with the various supervised schemes that mostly varied within 0.31 sec to 5 minute as reported in [18]. Some of our detected visual results on PETS2009 dataset are shown in the Fig. 7.

V. CONCLUSIONS

In this paper, we proposed a training-less model of pedestrian detection with no camera motion. This type of solution is quite useful as it is difficult to always guarantee availability of labelled training data, *e.g.*, surveillance in remote areas. As a part of our solution, we used an ellipse fitting model for accurate detection of pedestrians (within bounding boxes). A quality factor, based on area and entropy, is coined to perform discriminatory evaluation. Experimental results on PETS 2009 dataset clearly indicate that our approach compares quite well with many existing supervised approaches. In future, we will extend our present model to handle camera motion. Another direction of the research will be to better tackle the high density crowd scenarios.

VI. REFERENCES

- [1] P. Dollar, C. Wojek, B. Schiele, and P. Perona, "Pedestrian detection: An evaluation of the state of the art," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 34, no. 4, pp. 743–761, 2012.
- [2] J. Cao, Y. Pang, and X. Li, "Pedestrian detection inspired by appearance constancy and shape symmetry," *IEEE Trans. Image Process.*, vol. 25, no. 12, pp. 5538–5551, 2016.
- [3] B. Wu and R. Nevatia, "Detection of multiple, partially occluded humans in a single image by bayesian combination of edgelet part detectors," in *Proceedings of ICCV*, 2005, pp. 90–97.
- [4] Donghoon Lee, Geonho Cha, Ming-Hsuan Yang, and Songhwai Oh, "Individualness and determinantal point processes for pedestrian detection," in *Proceedings of ECCV*, 2016, pp. 330–346.
- [5] H. Idrees, K. Soomro, and M. Shah, "Detecting humans in dense crowds using locally-consistent scale prior and global occlusion reasoning," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 37, no. 10, pp. 1986–1998, 2015.
- [6] D. G. Lowe, "Distinctive image features from scale-invariant keypoints," *IJCV*, vol. 60, no. 2, pp. 91–110, 2004.
- [7] N. Dalal and B. Triggs, "Histograms of oriented gradients for human detection," in *Proceedings of CVPR*, 2005, pp. 886–893.
- [8] S. ul Hussain and B. Triggs, "Feature sets and dimensionality reduction for visual object detection," in *Proceedings of BMVC*, 2010, pp. 112.1–112.10.
- [9] P. Felzenszwalb, D. McAllester, and D. Ramanan, "A discriminatively trained, multiscale, deformable part model," in *Proceedings of CVPR*, 2008, pp. 1–8.
- [10] P. F. Felzenszwalb, R. B. Girshick, D. McAllester, and D. Ramanan, "Object detection with discriminatively trained part-based models," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 32, no. 9, pp. 1627–1645, 2010.
- [11] L. Wang, L. Xu, and M. H. Yang, "Pedestrian detection in crowded scenes via scale and occlusion analysis," in *Proceedings of ICIP*, 2016, pp. 1210–1214.
- [12] B. Leibe, E. Seemann, and B. Schiele, "Pedestrian detection in crowded scenes," in *Proceedings of CVPR*, 2005, vol. 1, pp. 878–885.
- [13] S. Agarwal and D. Roth, "Learning a sparse representation for object detection," in *Proceedings of ECCV*, 2002, pp. 113–127.
- [14] Z. Lin, G. Hua, and L.S. Davis, "Multiple instance feature for robust part-based object detection," in *Proceedings in IEEE CVPR*, 2009, pp. 405–412.
- [15] S. C. S. Cheung and C. Kamath, "Robust background subtraction with foreground validation for urban traffic video," *EURASIP Journal on Applied Signal Processing*, vol. 2005, no. 14, pp. 2330–2340, 2005.
- [16] S. Beucher and F. Meyer, "The morphological approach to segmentation: the watershed transformation. Mathematical morphology in image processing," *Optical Engineering*, vol. 34, pp. 433–481, 1993.
- [17] F. Meyer, "Topographic distance and watershed lines," *Signal Processing*, vol. 38, no. 1, pp. 113–125, 1994.
- [18] P. Dollar, R. Appel, S. Belongie, and P. Perona, "Fast feature pyramids for object detection," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 36, no. 8, pp. 1532–1545, 2014.
- [19] "PETS 2009 Benchmark Data," <http://www.cvg.reading.ac.uk/PETS2009/a.html>, March 2009.
- [20] Shaoqing Ren, Kaiming He, Ross Girshick, and Jian Sun, "Faster R-CNN: Towards real-time object detection with region proposal networks," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 39, no. 6, pp. 1137–1149, jun 2017.